

CSE331 Assignment 2 Part B

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1. Draw Push-down Automata for the following languages:
 - a. $L = \{0^i 1^j 0^k \mid i + j < k\}$
 - b. $L = \{a^n b^m \mid m, n > 0, n \neq m\}, \Sigma = \{a, b\}$
 - c. $L = \{a^n b^m \mid m, n > 0, n = 4m + 2\}, \Sigma = \{a, b\}$
 - d. $L = \{a^n b^m c^p \mid m, n, p > 0, m=n \text{ or } m=p\}$
 - e. $L = \{w_1 \# w_2 \mid \text{the number of } 01 \text{ in } w_1 \text{ is equal to the number of } 10 \text{ in } w_2\}, \Sigma = \{0, 1\}$
2. Draw Turing Machine for the following languages:
 - a. Given an input string w over the alphabet $\{a, b\}$, writes the same string again after a delimiter symbol $\#$. The TM should halt with the tape containing $w\#w$, where the first occurrence of w is the input and the second is its copy.
 - b. $L = \{w \in \{0, 1\}^* \mid w \text{ is a palindrome}\}$
 - c. $L = \{a^i b^j c^k \mid i * j = k\}$
3. Prove that:
 - a. The collection of decidable languages is closed under Union
 - b. The collection of decidable languages is closed under Complement
4. Prove using reduction that:
 - a. A_{TM} is undecidable
 - b. E_{TM} is undecidable